PRANAV KETHIREDDY

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PROFILE SUMMARY

2nd-year Electrical Engineering undergraduate at IIT Delhi with a strong foundation in data structures, algorithms, and systems programming. Built a Redis-compatible server from first principles (async I/O, RESP protocol, AOF persistence) and a Codeforces analytics dashboard in vanilla JavaScript, and has solved 130+ competitive programming problems. Seeking an internship at the intersection of algorithms and systems.

EDUCATION

Indian Institute of Technology (IIT) Delhi — B.Tech in Electrical Engineering 2025 – 2029 (Expected)
Class XII (BIEAP) 98.2%
Class X (CBSE) 92.2%
Relevant Coursework: Calculus, Linear Algebra & Differential Equations, Introduction to Probability & Stochastic Processes, Python Programming

PROJECTS

Mini Redis — Redis-Compatible Key-Value Server *(Python, asyncio)* GitHub

- Built a Redis-compatible in-memory key-value server from scratch supporting strings, hashes and TTL expiry (14 commands), compatible with *redis-cli* and standard Redis client libraries.
- Hand-wrote an incremental RESP protocol parser and encoder that handles partial TCP reads, pipelined commands and malformed input.
- Used a single-threaded asyncio event loop to serve concurrent clients without locks, mirroring Redis's own concurrency model.
- Implemented append-only-file (AOF) persistence with fsync and crash-safe replay, logging expiries as absolute timestamps so TTLs stay correct across restarts.
- Wrote 55+ unit and integration tests across the store, protocol, dispatcher and TCP server, uncovering 8 bugs fixed through test-driven, one-fix-per-commit changes.

Codeforces Analytics Dashboard *(HTML, CSS, JavaScript)* GitHub

- Built a client-side analytics dashboard in vanilla JavaScript, with no framework or backend, consuming the live Codeforces REST API (user.info, user.status, user.rating, contest.list).
- Computed profile analytics (problems solved, rating history, maximum rating change) from raw submission and rating data, and added upcoming-contest tracking and head-to-head user comparison.
- Persisted saved handles, bookmarks, goals and theme in localStorage, and designed a responsive tabbed UI with dark/light themes, loading skeletons and API error handling.

ACADEMIC ACHIEVEMENTS

- Secured 99.91 percentile in JEE Main and AIR 2271 in JEE Advanced (2025), among the top ~0.1% of ~1 million candidates.
- Qualified for Round 2 of the Regional Mathematics Olympiad (RMO).
- Ranked 1st out of 250+ students at the school level in the Ramanujan Mathematics Olympiad.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++ (basics), HTML/CSS
Libraries/Tools: asyncio, pytest, Git, NumPy, Pandas, REST APIs, basic Machine Learning concepts
Core Competencies: Data Structures & Algorithms, Systems Programming, Network Protocols, Object-Oriented Programming, Competitive Programming

COMPETITIVE PROGRAMMING PROFILE

- Codeforces rating: 1201 (Pupil).
- Solved 130+ problems across Codeforces, CodeChef and LeetCode covering arrays, graphs, dynamic programming and binary search.
- Actively training on graph algorithms, dynamic programming and binary search on answer for upcoming contests and hiring challenges.

ADDITIONAL INFORMATION

- Touch-typing speed: 121+ WPM.